

A0_2421583

June 6, 2026

```
[1]: # import built-in libraries
from pathlib import Path

# import third-party libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: data_path = Path("./projects/data/customer_churn.csv")
df = pd.read_csv(data_path)
df.head(3)
```

```
[2]: Customer ID  Gender  Age  Married  Number of Dependents  City \
0  0002-ORFBO  Female  37      Yes              0  Frazier Park
1  0003-MKNFE   Male  46      No              0    Glendale
2  0004-TLHLJ   Male  50      No              0    Costa Mesa

   Zip Code  Latitude  Longitude  Number of Referrals  ...  Payment Method \
0    93225  34.827662 -118.999073              2  ...    Credit Card
1    91206  34.162515 -118.203869              0  ...    Credit Card
2    92627  33.645672 -117.922613              0  ...  Bank Withdrawal

   Monthly Charge Total Charges  Total Refunds  Total Extra Data Charges  \
0              65.6          593.30           0.00              0
1              -4.0          542.40          38.33             10
2              73.9          280.85           0.00              0

   Total Long Distance Charges  Total Revenue  Customer Status  Churn Category  \
0              381.51          974.81        Stayed           NaN
1              96.21          610.28        Stayed           NaN
2              134.60          415.45        Churned    Competitor

   Churn Reason
0           NaN
1           NaN
2  Competitor had better devices
```

[3 rows x 38 columns]

```
[3]: pd.set_option('display.max_rows', None)
df.columns
```

```
[3]: Index(['Customer ID', 'Gender', 'Age', 'Married', 'Number of Dependents',
        'City', 'Zip Code', 'Latitude', 'Longitude', 'Number of Referrals',
        'Tenure in Months', 'Offer', 'Phone Service',
        'Avg Monthly Long Distance Charges', 'Multiple Lines',
        'Internet Service', 'Internet Type', 'Avg Monthly GB Download',
        'Online Security', 'Online Backup', 'Device Protection Plan',
        'Premium Tech Support', 'Streaming TV', 'Streaming Movies',
        'Streaming Music', 'Unlimited Data', 'Contract', 'Paperless Billing',
        'Payment Method', 'Monthly Charge', 'Total Charges', 'Total Refunds',
        'Total Extra Data Charges', 'Total Long Distance Charges',
        'Total Revenue', 'Customer Status', 'Churn Category', 'Churn Reason'],
        dtype='str')
```

```
[4]: pd.concat([df.dtypes, df.isnull().sum()], axis=1).T
```

```
[4]: Customer ID Gender      Age Married Number of Dependents City Zip Code \
0          str      str  int64      str              int64  str  int64
1          0      0      0      0              0      0      0

      Latitude Longitude Number of Referrals ... Payment Method Monthly Charge \
0  float64  float64              int64 ...      str      float64
1          0          0              0 ...      0          0

      Total Charges Total Refunds Total Extra Data Charges \
0  float64      float64              int64
1          0          0              0

      Total Long Distance Charges Total Revenue Customer Status Churn Category \
0              float64      float64      str      str
1              0          0          0      3488

      Churn Reason
0          str
1      3488
```

[2 rows x 38 columns]

```
[5]: # df[['Monthly Charge']].value_counts()
```

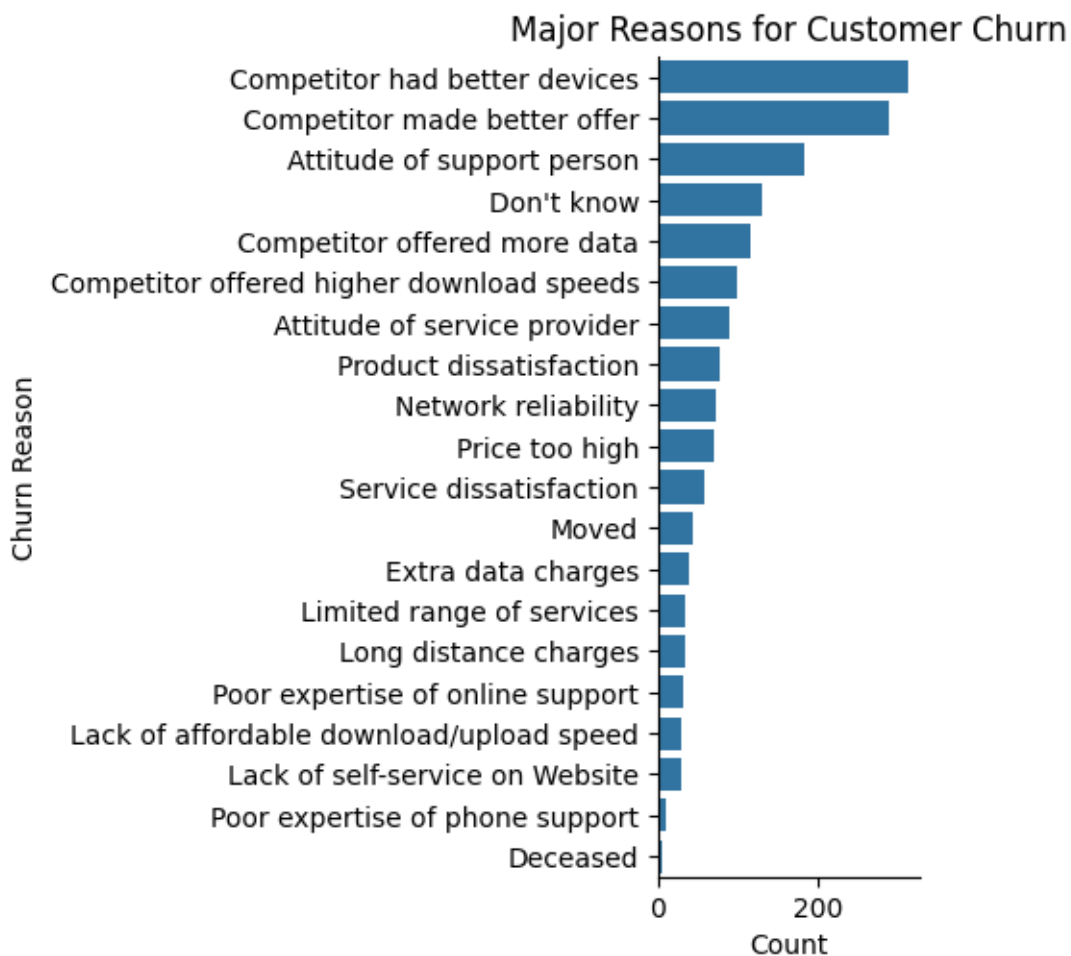
QA_1: What is the customer churn rate? Determine the total revenue lost from churning? Determine the total revenue lost from churning?

```
[6]: churn_rate = df['Customer Status'].value_counts(normalize=True).loc['Churned']
print("customer churn rate:", churn_rate)
total_revenue_churned = df.loc[df['Customer Status'] == 'Churned', 'Total_Revenue'].sum()
print("total revenue from churned customers:", total_revenue_churned)
```

customer churn rate: 0.3348588863463005
total revenue from churned customers: 3641352.63

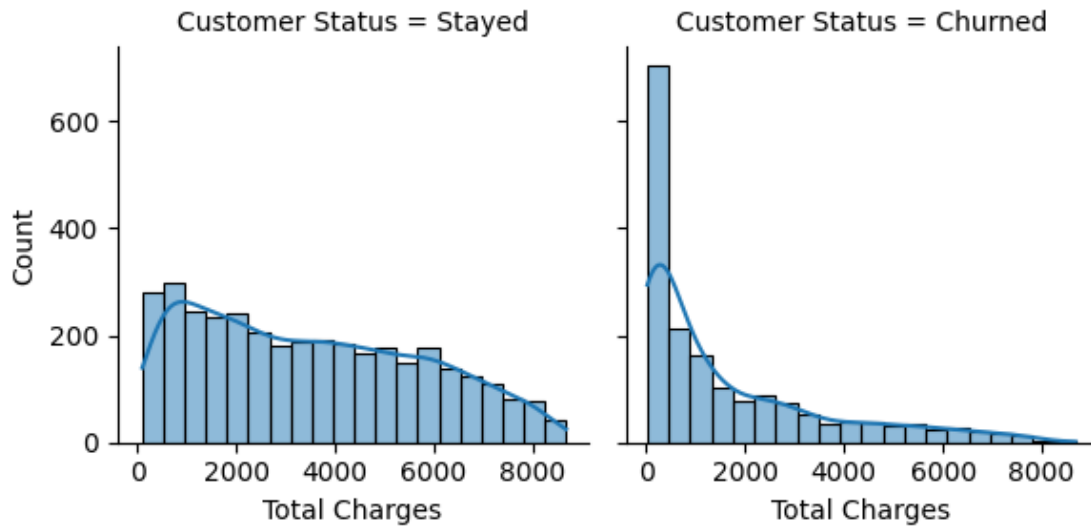
QA_2:Strellis bar chart. how the major reasons for churning using a

```
[7]: # df["Churn Reason"].value_counts().sort_values(ascending=False).plot.bar()
order = df.query("`Customer Status` == 'Churned'")["Churn Reason"].
value_counts().index
sns.catplot(data=df.query("`Customer Status` == 'Churned'"), y="Churn Reason",
kind="count", order=order )
plt.title("Major Reasons for Customer Churn")
plt.ylabel("Churn Reason")
plt.xlabel("Count")
plt.show()
```

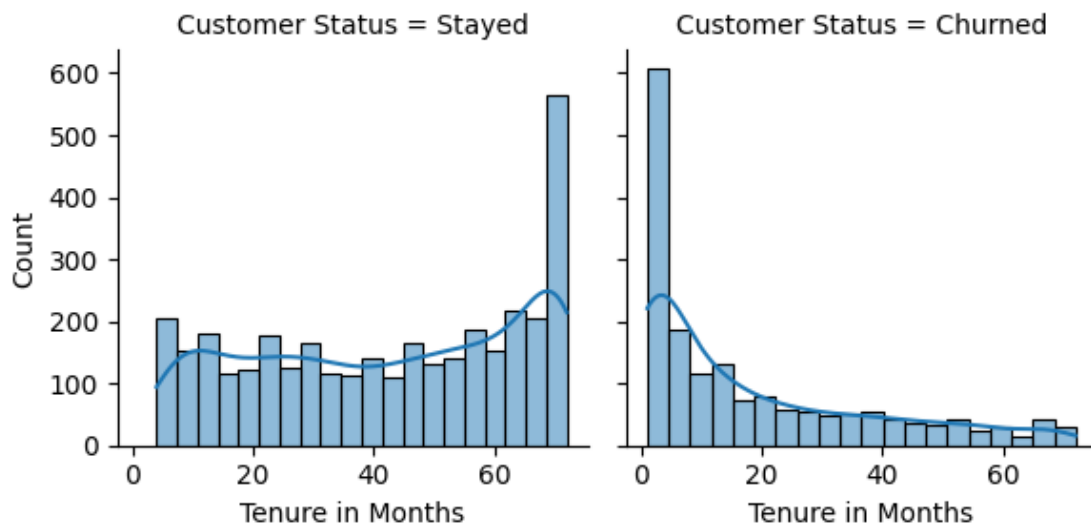


QA_3: Discuss the distribution of customers that stayed and churned for the selected variables (total_charges, revenue, and tenure_in_months) using a histogram.

```
[8]: g = sns.FacetGrid(df, col="Customer Status",)
g.map(sns.histplot, "Total Charges", kde=True, bins=20)
plt.show()
```

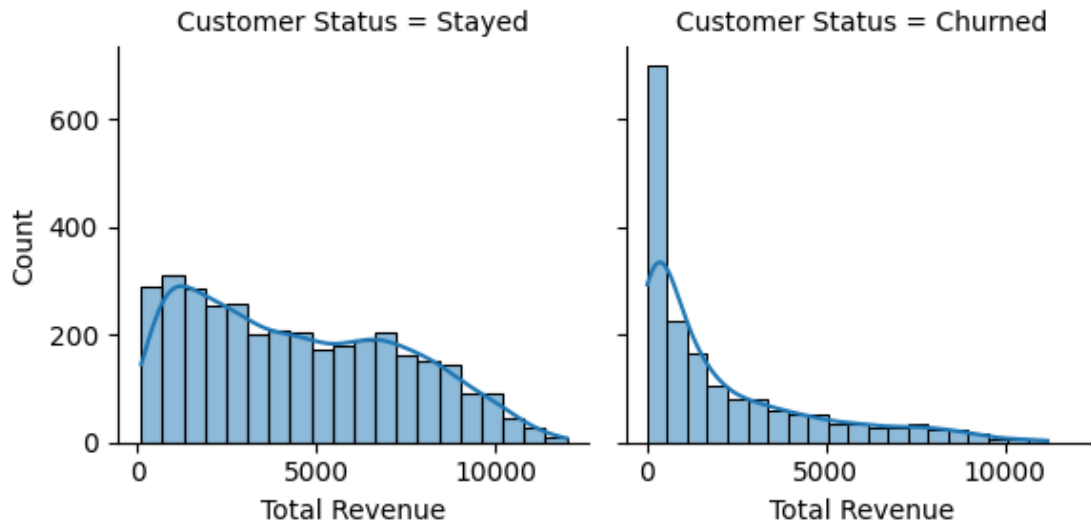


```
[9]: g = sns.FacetGrid(df, col="Customer Status")
g.map(sns.histplot, "Tenure in Months", kde=True, bins=20)
plt.show()
```



```
[10]: g =sns.FacetGrid(df, col="Customer Status")
g.map(sns.histplot, "Total Revenue", kde=True, bins=20)
```

```
[10]: <seaborn.axisgrid.FacetGrid at 0x120fe5550>
```



The histograms show clear differences between stayed and churned customers.

- For Total Charges, churned customers are concentrated at lower values, while stayed customers have a wider distribution and more customers with high total charges. This suggests that customers who churned usually spent less overall.
- For Total Revenue, churned customers also tend to have lower total revenue compared with stayed customers. Stayed customers contribute more revenue because they usually remain with the company longer.
- For Tenure in Months, churned customers are mostly concentrated in shorter tenures, especially in the first months. Stayed customers have a broader distribution and many have much longer tenure. This indicates that customers with shorter tenure are more likely to churn.

QA_4: Complete the following using a pivot table. i. Determine the total revenue contribution from a Two-Year contract for each Offer by internet type. ii. For Offer B, what (%) of the total revenue was contributed by churned customers that accepted a Month-to-Month contract for Cable service.

```
[11]: #i
pt= df.pivot_table(index=["Offer",], columns=["Contract","Internet Type"],
    ↪ values="Total Revenue", aggfunc="sum")
pt.loc[:,("Two Year",)]
```

```
[11]: Internet Type      Cable      DSL  Fiber Optic
Offer
Offer A      349750.71  720759.09  1474862.52
Offer B      164188.21  362671.78  473681.65
Offer C       10360.26  34148.92   80196.94
Offer D        5443.22   5592.51   22464.25
Offer E        1684.86   3019.54    6134.10
```

```
[12]: #ii
pt = df.pivot_table(index=["Offer","Contract","Internet Type"],
    ↪columns="Customer Status", values="Total Revenue",
    ↪aggfunc="sum",fill_value=0)
# pt
```

```
[13]: num = pt.loc[("Offer B", "Month-to-Month", "Cable"),"Churned"]
num
denom = pt.loc["Offer B"].sum().sum()
denom
churn_rate_offer_b = num / denom *100
print(f"For Offer B, Month-to-Month, Cable: {churn_rate_offer_b:.2f}%")
```

For Offer B, Month-to-Month, Cable: 0.45%